Analysis of LendingClub Portfolio Performance in 2018

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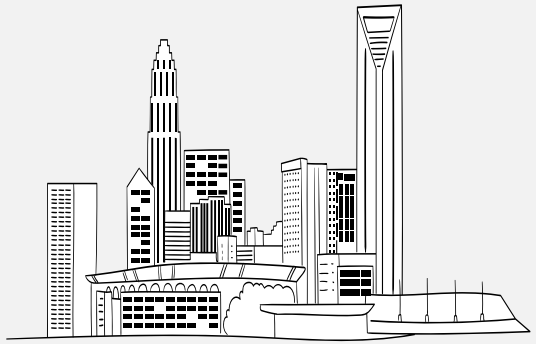
Outline



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Executive Summary

- In 2018, LendingClub funded a total of 57,160 loans amounting to $1,138.2M.
- 12.6% of loans were charged off; total losses amounted to $90.8M.
- Average interest rate was 14.75%; DTI ratio averaged 20.97%.
- Key insights revealed trends and disparities in lending performance impacting profitability and risk management.

Introduction

- The objective of this report is to analyze LendingClub's 2018 loan data reviewing key metrics and performance trends such as, total loan metrics, quality of loans, borrower profiles, regional performance, and financial health.
- The significance of understanding lending performance is to inform strategic decisions for improved profitability and operational efficiency.

Methodology

- Data was extracted from a subset of the year 2018 of a real-world multi-year dataset sourced from LendingClub, including geographic details, borrower demographics, and repayment behavior metrics

- Analytical Techniques:
  - Descriptive statistics for trends and performance metrics.
  - Comparative analysis for loan status (Good vs. Bad).
  - Contingency table to explore borrower profiles and repayment patterns.

- Tools Used:
  - Database software for data analysis (SQL).
  - Visualization software for dashboard (Tableau).

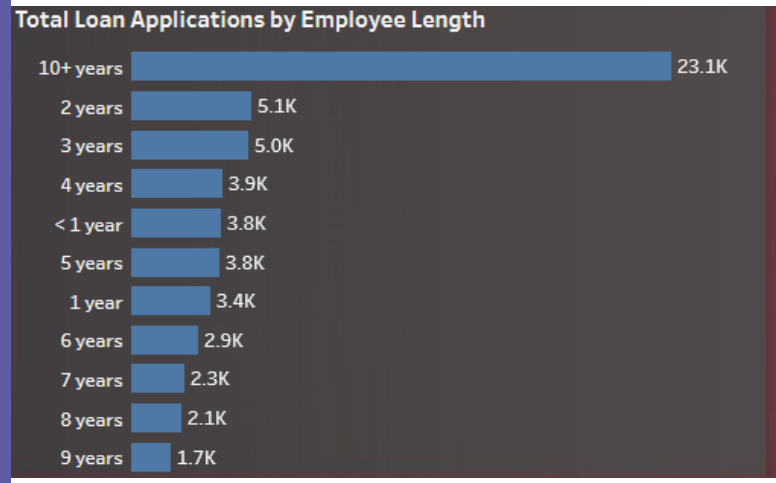
Results

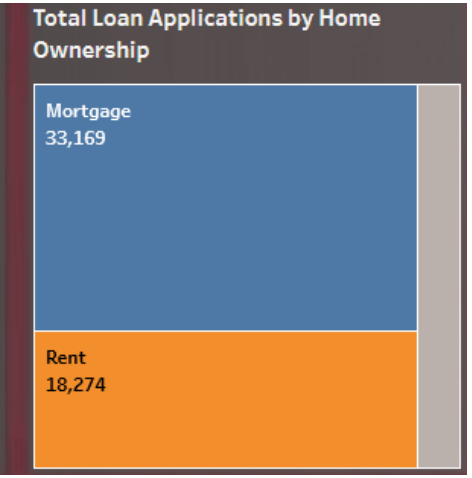
Total Loans and Financials

- 57,160 loans issued at an average interest rate of 14.7% in 2018
- Loaned $1,138.2M total to approved loan applicants
- Payments received totaled $840M on fully paid, current, and charged off loans.

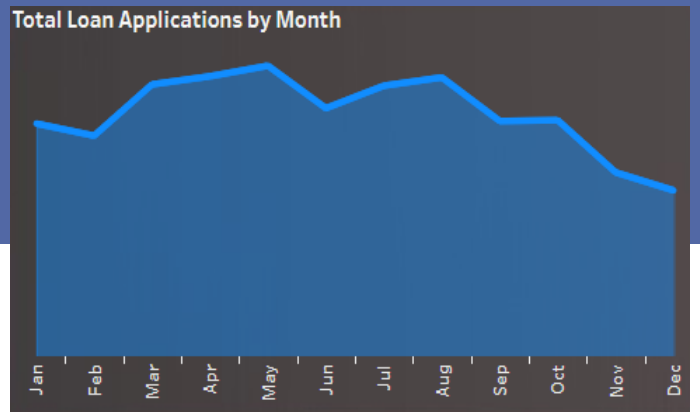
Loan Performance and Borrower Profiles

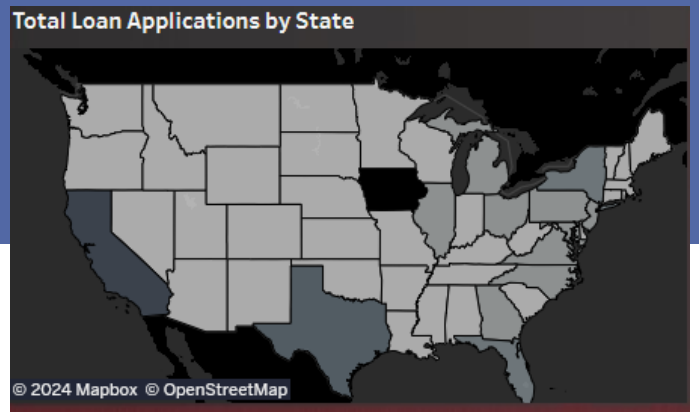
- 87.4% of loans are current or fully paid; 12.6% charged off equating to $90.8 million loss.
- Grading criteria reveals 38% of borrowers are A and B (low risk), while 62% are C – G (moderate to very high risk).
- Employment history shows 40% of borrowers had 10+ years of experience; 58% had mortgages.
- Borrower inclination towards shorter-term (3 Yr) vs medium-term (5 Yr) loans





Findings & Implications





Findings

- Applications peaked in May; December saw the least activity.
- Grand majority (80%) of loans were for debt consolidation and credit cards
- California, Texas, and Florida are top states for loans; opportunity identified in underserved states.
- Profit on fully paid loans (12%) is less than interest rates (14.6%); charge-offs present major losses (-62.5%).

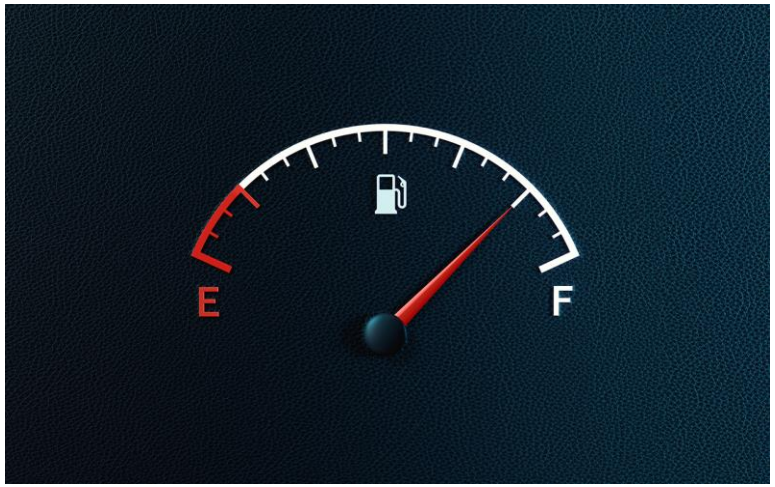
Implications

- Strategic focus on early-year lending opportunities could drive growth
- Borrowers in these categories may be facing financial difficulties and these loan purposes may increase lending risk despite manageable DTI levels
- Targeted marketing strategies in high-performing and underserved regions can enhance loan distribution.
- Refining risk assessments and redesigning loan offerings to minimize defaults is essential.

Dashboard

2018 Lending Club Loan Report

Tableau Dashboard



https://public.tableau.com/app/profile/alexander.porter3156/viz/2018LoanReport/Summary

2018 LendingClub Loan Report | SUMMARY

Total Loan Applications	Total Funded Amount	Total Amount Received	Average Interest Rate	Average Debt to Income
57.2K	$1,138.2M	$840.0M	14.7%	21.0%
MTD 3.3K	MTD $65.0M	MTD $38.5M	MTD 14.3%	MTD 20.8%
MoM -9.79%	MoM -12.2%	MoM -15.4%	MoM 0.3%	MoM -0.2%

Purpose
(All)

State
(All)

Grade
(All)

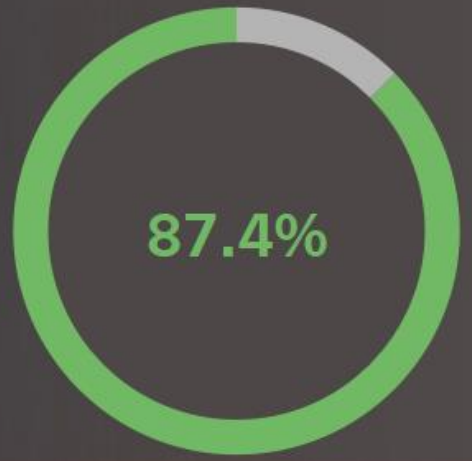
BAD LOANS ISSUED

Applications
7.2K

Funded Amount
$145.3M

Amount Received
$54.5M

GOOD LOANS ISSUED



Applications
50.0K

Funded Amount
$992.9M

Amount Received
$785.5M

Summary

Overview

Details

LOAN STATUS

Loan Status	Total Loan Applications	Total Funded Amount	Total Amount Received	MTD Total Amount Received	MTD Total Funded Amount	Average Interest Rate	Average Debt to Income
Charged Off	7.2K	$145.34M	$54.52M	$1.61M	$6.18M	17.4%	21.3%
Current	32.5K	$668.94M	$422.60M	$21.94M	$45.21M	14.2%	21.1%
Fully Paid	17.5K	$323.93M	$362.90M	$14.95M	$13.65M	14.6%	20.7%
Grand Total	57.2K	$1,138.20M	$840.02M	$38.51M	$65.05M	14.7%	21.0%

2018 LendingClub Loan Report | OVERVIEW

Total Loan Applications	Total Funded Amount	Total Amount Received	Average Interest Rate	Average Debt to Income
57.2K	$1,138.2M	$840.0M	14.7%	21.0%
MTD 3.3K	MTD $65.0M	MTD $38.5M	MTD 14.3%	MTD 20.8%
MoM -9.79%	MoM -12.2%	MoM -15.4%	MoM 0.3%	MoM -0.2%

Select KPI
Total Loan Applications

State
(All)

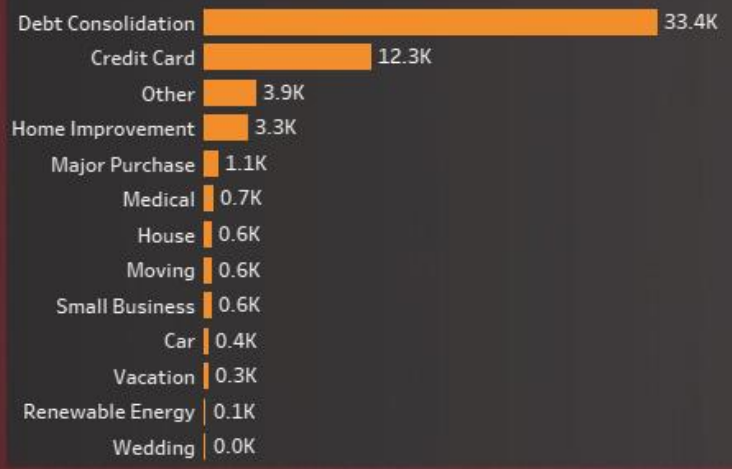
Grade
(All)

Summary

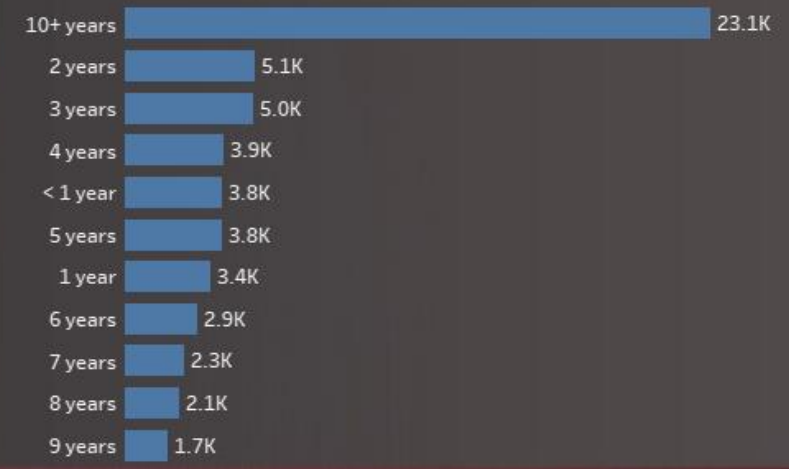
Overview

Details

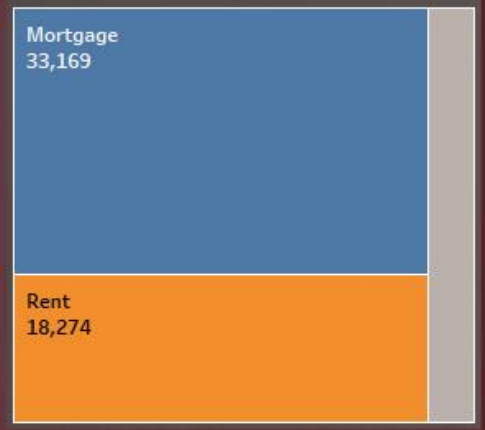
Total Loan Applications by Purpose



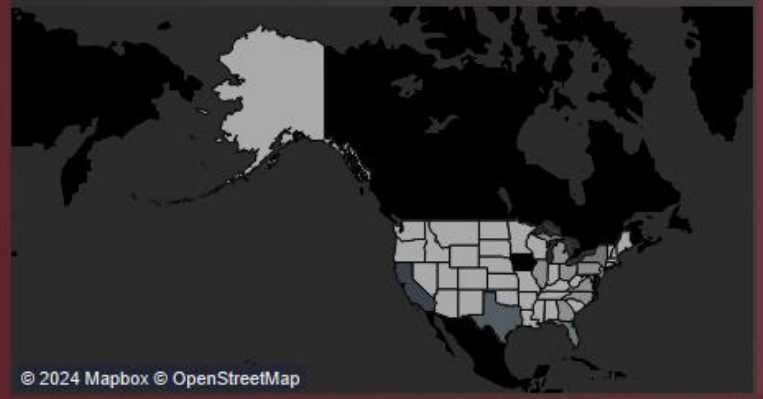
Total Loan Applications by Employee Length



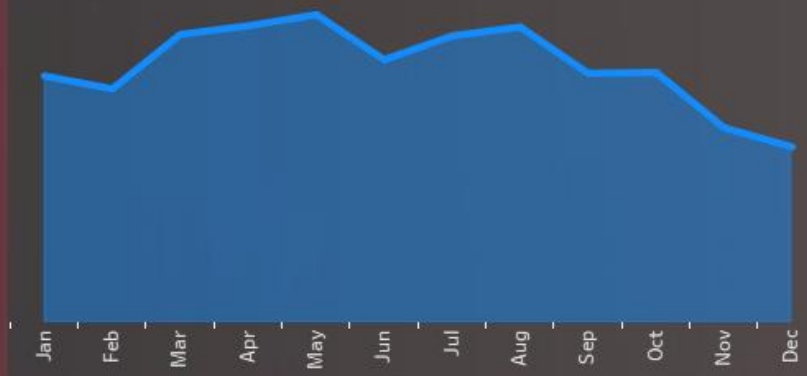
Total Loan Applications by Home Ownership



Total Loan Applications by State



Total Loan Applications by Month



Total Loan Applications by Term

2018 LendingClub Loan Report | DETAILS

Total Loan Applications		Total Funded Amount		Total Amount Received		Average Interest Rate		Average Debt to Income	
57.2K		$1,138.2M		$840.0M		14.7%		21.0%	
MTD	MoM	MTD	MoM	MTD	MoM	MTD	MoM	MTD	MoM
3.3K	-9.79%	$65.0M	-12.2%	$38.5M	-15.4%	14.3%	0.3%	20.8%	-0.2%

Purpose
(All)

Home Ownership
(All)

Grade
(All)

Sub Grade
(All)

Summary

Overview

Details

Id	Purpose	Home Ownership	Grade	Sub Grade	Month of Issue Date	Loan Amount	Interest Rate	Installment	Total Payment
118187703	Credit Card	Mortgage	A	A3	January	$40,000.00	6.7%	$1,229.97	$34,424.23
119987870	Debt Consolidation	Rent	D	D3	February	$11,200.00	19.0%	$290.72	$12,192.69
120068032	Debt Consolidation	Mortgage	D	D2	January	$20,475.00	18.1%	$520.60	$14,556.26
120640856	Small Business	Mortgage	A	A1	January	$40,000.00	5.3%	$1,204.60	$41,776.56
121263466	Debt Consolidation	Mortgage	A	A5	January	$25,000.00	8.0%	$783.07	$27,557.92
121703857	Debt Consolidation	Mortgage	C	C4	April	$30,000.00	15.0%	$714.33	$18,887.17
121785745	Other	Rent	C	C4	January	$6,000.00	15.1%	$208.14	$5,819.82
121919076	Other	Rent	B	B3	January	$26,350.00	10.4%	$565.33	$15,798.73
122265432	Debt Consolidation	Mortgage	C	C4	March	$40,000.00	15.0%	$952.44	$22,724.88
122298656	Credit Card	Mortgage	E	E3	January	$29,400.00	24.9%	$860.35	$31,544.57
122299236	Debt Consolidation	Own	B	B5	January	$36,000.00	12.0%	$800.62	$20,756.17
122359743	Other	Mortgage	B	B5	January	$7,200.00	12.0%	$239.11	$3,583.74
122662255	Credit Card	Mortgage	C	C3	February	$35,000.00	14.1%	$1,197.58	$40,618.24
123055362	Credit Card	Rent	E	E4	January	$10,000.00	25.8%	$298.34	$8,339.18
123615772	Home Improvement	Mortgage	C	C2	January	$20,000.00	13.6%	$461.13	$12,896.54
123882130	Debt Consolidation	Mortgage	A	A3	January	$40,000.00	6.7%	$1,229.97	$32,992.27
123894756	Debt Consolidation	Mortgage	C	C5	January	[illegible]	[illegible]	[illegible]	[illegible]

Conclusion

- Charge-offs predominantly arise from C-G graded loans, highlighting a critical area for risk assessment.
- The first half of the year is more favorable for loan issuance, while trends suggest caution in the latter half.
- Need for Enhanced Risk Models: Focus on improving assessment mechanisms for grading and potential charge-off predictors.
- The analysis reveals significant issues with loan performance and a minimum net loss of approximately 3.9% is expected for 2018

Grade	Total_ChargeOffs	Total_Loans_%	Avg_Interest_Rate
A	332	0.5808	0.0735
B	1007	1.7617	0.1109
C	2274	3.9783	0.1481
D	2412	4.2197	0.1971
E	900	1.5745	0.2546
F	204	0.3569	0.2948
G	48	0.0840	0.3082

Actionable Insights for LendingClub

Refine risk management and implement stricter criteria for C-G graded borrowers to curb charge offs

Consider restructuring underwriting to attract and approve only A and B graded borrowers to minimize risk and ensure net profitability.

Develop targeted marketing strategies during peak lending periods, particularly in high-performing states.

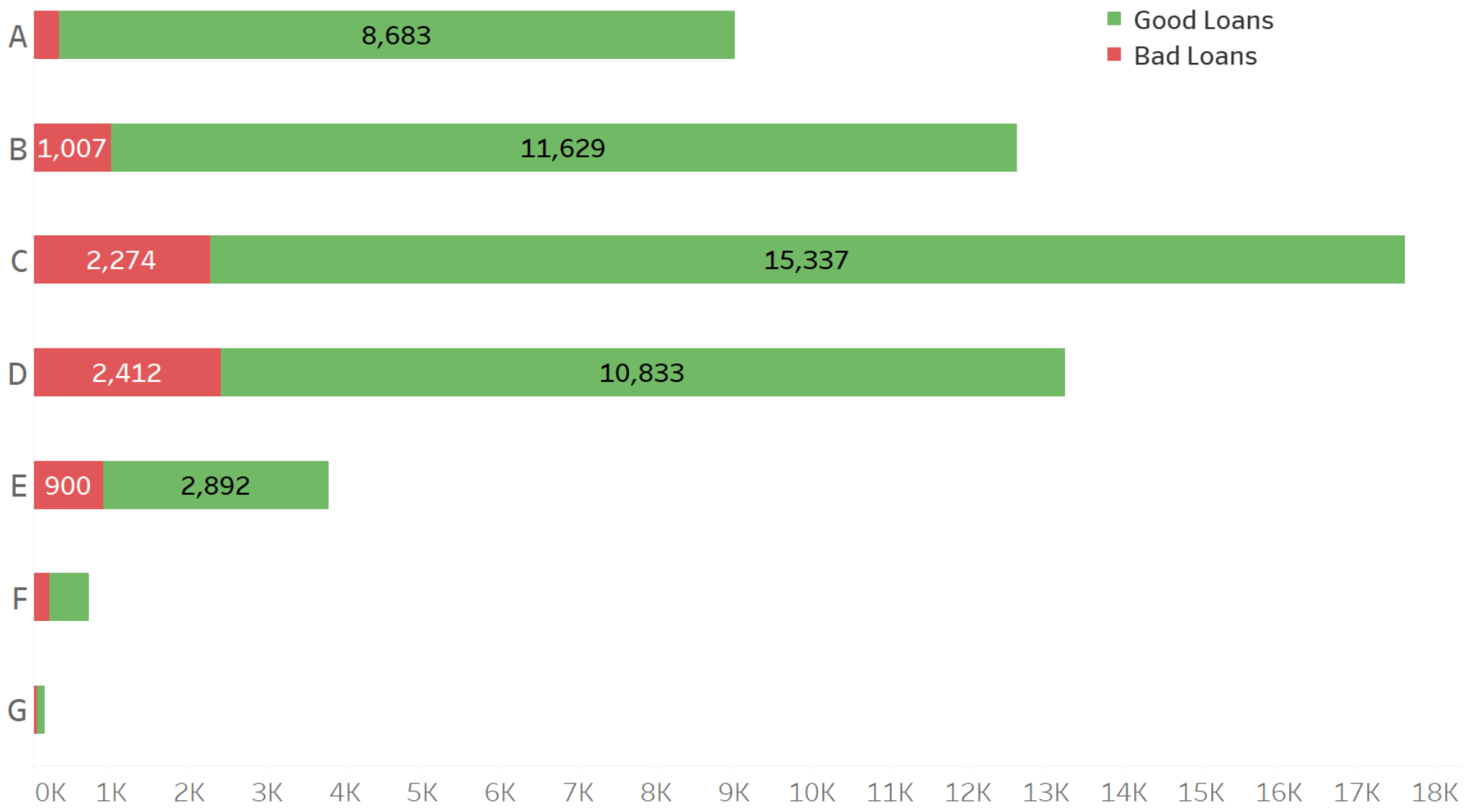
Explore opportunities for diversification into underserved regions and marginalized loan purposes to capture market share and reduce risk.

Appendix

Profit: Fully Paid vs. Charge-offs

Loan_Status	Profit_Multiple	Profit
Fully Paid	0.1203	$38,973,290
Charged Off	-0.6249	-$90,820,023

Distribution of Loan Status by Grade



Contingency Table

		Loan_Status (grouped)		
		Bad_Loans	Good_Loans	Total
Grade	A	332 (0.58%)	8683 (15.19%)	9015 (15.77%)
	B	1007 (1.76%)	11629 (20.34%)	12636 (22.11%)
	C	2274 (3.98%)	15337 (26.83%)	17611 (30.81%)
	D	2412 (4.22%)	10833 (18.95%)	13245 (23.17%)
	E	900 (1.57%)	2892 (5.06%)	3792 (6.63%)
	F	204 (0.36%)	513 (0.90%)	717 (1.25%)
	G	48 (0.08%)	96 (0.17%)	144 (0.25%)
	Total	7177 (12.56%)	49983 (87.44%)	57160 (100.00%)